

SMARTGEAR RETAIL 2.0 ASSIGNMENT

Instructions to Students

- This is an individual assignment.
- Use Databricks Community Edition and Confluent Cloud free tier.
- Use Azure concepts where required (limited practical implementation expected).
- Submit all deliverables, including notebooks, code, screenshots, and theory answers.
- Clearly document assumptions wherever required.

Dataset Strategy & Data Sources

This assignment uses a combination of **batch, streaming, and semantic data sources** to simulate a real-world modern data platform.

1. Batch Dataset (Provided)

You will reuse the dataset from Assignment 1:

- **File Name:** smartgear_sales.csv
- This dataset represents historical sales data.

Usage:

- Batch processing
- Validation of streaming outputs
- Synapse analytics
- Benchmarking and reconciliation

2. Streaming Dataset (To Be Created by Students)

You are required to simulate real-time data using Kafka.

Schema to be used:

- order_id
- timestamp
- store_id
- product
- quantity
- price
- region

Usage:

- Kafka ingestion
- Real-time processing in Databricks
- Bronze → Silver → Gold pipeline

3. Product Metadata Dataset (To Be Created by Students)

You must create a small dataset for semantic search use case.

Example Schema:

- product_id
- product_name
- category
- description

Usage:

- Embedding generation
- Vector database storage
- Semantic search implementation

Important Note

*This assignment is designed to simulate a **hybrid data platform**, where:*

- *Batch and streaming pipelines coexist*
- *Analytical and real-time systems are integrated*
- *Structured and unstructured data are both leveraged*

*Students are expected to **connect these datasets logically** as part of their solution.*

PART A – Architecture Thinking (20 Marks)

Q1. Design End-to-End Data Platform (10 Marks)

Draw and explain architecture covering:

- Kafka ingestion layer
- Databricks streaming + batch processing
- Medallion architecture
- Delta Lake usage
- Synapse analytics
- Vector search layer

Explain:

- Why Lakehouse over traditional DW?
- Where batch vs streaming fits?
- Where cost trade-offs arise?

Q2. Databricks vs Synapse (5 Marks)

Compare:

- Spark in Databricks vs Synapse Spark
- SQL Pools vs Databricks SQL
- When to use each in an enterprise setup

Q3. Delta Lake Fundamentals (5 Marks)

Explain:

- ACID in data lakes
- Schema evolution
- Time travel

- Why Delta is critical for streaming pipelines

PART B – Kafka + Streaming Pipeline (25 Marks)

B1 – Kafka Theory (10 Marks)

Answer clearly:

1. What is a Topic, Partition, Offset?
2. How does Kafka guarantee durability?
3. Difference between:
 - At-most once
 - At-least once
 - Exactly-once
4. What is consumer group rebalancing?
5. Why partitioning matters for scalability?

B2 – Synthetic Streaming Data Generator (5 Marks)

You must:

- Create Python code to simulate streaming events:
order_id, timestamp, store_id, product, quantity, price, region

Requirements:

- Randomized realistic values
- Continuous stream simulation
- Batch push every few seconds

B3 – Kafka Implementation (5 Marks)

Using Confluent Cloud:

1. Create topic: smartgear_orders
2. Configure:
 - Partitions (justify choice)
 - Retention
3. Implement:
 - Producer (your generator)
 - Consumer (basic validation)

B4 – Streaming to Databricks (5 Marks)

Using Spark Structured Streaming:

- Read from Kafka
- Parse JSON
- Write to Bronze layer (Delta)

Must include:

- Checkpointing
- Schema enforcement

PART C – Lakehouse & Medallion (25 Marks)

C1 – Bronze Layer (5 Marks)

Using Databricks:

- Ingest streaming data from Kafka
- Store as append-only Delta table
- Add metadata:
 - ingestion_time
 - source
- Ensure schema enforcement

C2 – Silver Layer (10 Marks)

Data Engineering Tasks (5 Marks)

Using Spark (PySpark or SQL):

- Handle nulls:
 - Drop or impute (justify choice)
- Standardize:
 - Region values (case normalization, mapping)
- Remove duplicates:
 - Based on order_id + timestamp
- Create:
 - revenue = quantity * price
- Ensure correct data types

Analytical Transformations (5 Marks)

Using Spark SQL OR DataFrame APIs, perform:

1. Daily Revenue Trend
 - Aggregate revenue by date
 - Sort chronologically
2. Region-wise Contribution %
 - % contribution of each region to total revenue
3. Top 3 Stores per Region
 - Use window functions
 - Rank stores within each region
4. Data Skew Identification
 - Identify if any region/store dominates data disproportionately

C3 – Gold Layer (10 Marks)

Executive-ready analytics + advanced SQL

Create final curated datasets:

Business KPIs (5 Marks)

Using Spark SQL:

1. Region-wise total revenue
2. Month-over-month growth (MoM)
3. Average order value per region
4. Store performance segmentation:
 - High / Medium / Low (based on revenue percentile)

Advanced SQL Analytics (5 Marks)

Students must use window functions + advanced SQL

1. Top 5 Products by Revenue (Global + Region-wise)
 - Use RANK() / DENSE_RANK()
2. Rolling Revenue (7-day moving average)
 - Use window frame
3. Anomaly Detection (Basic)
 - Identify days where revenue deviates >30% from average
4. Customer Behavior Proxy (Optional Challenge Bonus)
(since no customer ID – force thinking)
 - Use StoreID as proxy → identify “repeat behavior patterns”

PART D – Orchestration (10 Marks)

D1 – Airflow Theory (5 Marks)

Explain:

- DAG structure
- Idempotency
- Backfills
- Retry mechanisms
- Sensors

D2 – ADF Pipeline Design (5 Marks)

Design (no need to implement fully):

- Pipeline for:
 - Kafka → Bronze → Silver → Gold
- Include:
 - Trigger strategy
 - Monitoring
 - Cost considerations

PART E – Streaming Design Patterns (5 Marks)

Answer:

1. Stateful vs Stateless streaming

2. What is watermarking?
3. Handling late-arriving data
4. Exactly-once in Spark + Kafka (how?)

PART F – Vector DB & Semantic Search (10 Marks)

F1 – Theory (5 Marks)

Explain:

- What are embeddings?
- Why cosine similarity works?
- Difference between keyword search vs semantic search
- What is ANN (Approximate Nearest Neighbour)?

F2 – Practical (5 Marks)

Using Databricks + local setup:

1. Create product descriptions dataset
2. Generate embeddings
3. Store in vector DB (Chroma or similar)
4. Implement semantic search function

PART G – Mini Case (5 Marks)

Scenario:

SmartGear leadership observes:

- Revenue mismatch between streaming and batch reports
- Sudden spike in Kafka lag
- Missing records in Gold layer

Question:

As a Data Engineering Lead:

1. What could be the root causes?
2. How would you debug?
3. What monitoring would you design?

Deliverables

Students must submit:

1. Databricks notebooks (Bronze/Silver/Gold + Streaming)
2. Kafka producer + consumer code
3. Architecture diagram (mandatory)
4. Synapse SQL scripts
5. Vector search implementation
6. Theory answers document
7. Screenshots:
 - Kafka topics
 - Streaming output

- Delta tables
- 8. Reflection (200 words):
 - Cost vs performance trade-offs

Evaluation Philosophy

This assignment evaluates:

- System thinking (not just coding)
- Understanding of streaming vs batch
- Ability to design production systems
- Trade-off awareness (cost, latency, scalability)
- End-to-end ownership mindset

